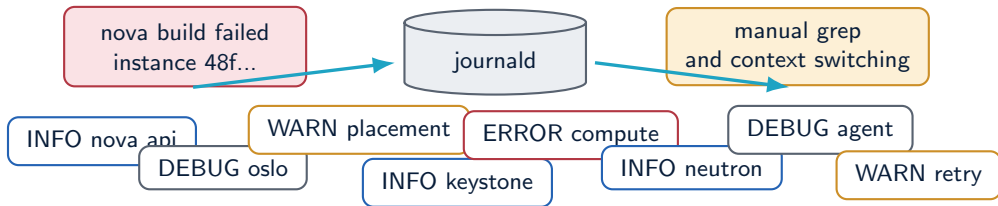


OpenStack RCA Copilot

Noisy cloud logs become structured, bounded, explainable investigation evidence.

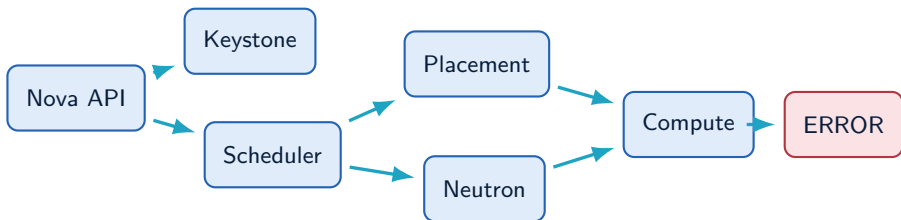
Logs → Events → Graph → Incident Evidence → Horizon → Grounded AI

1. Hook: Evidence Overload



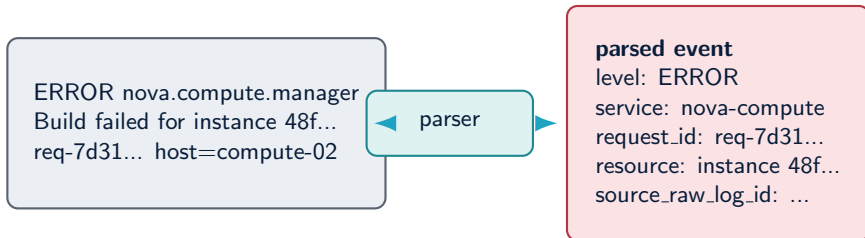
The failure is visible somewhere in the logs.
The hard part is finding the evidence around it.

2. Problem: Logs Are Noisy, Distributed, and Hard to Connect



One user operation crosses services, clocks, request IDs, resources, retry loops, and background noise. Raw logs alone show fragments, not an investigation path.

3. Insight: Every Log Can Become a Structured Event



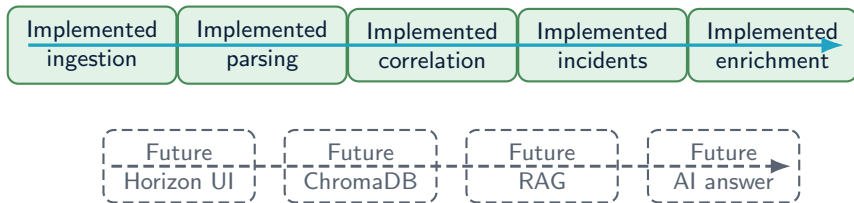
The system does not discard the original record. It adds structure and lineage so evidence can be queried, linked, and audited.

4. Pipeline Overview: The Mental Model



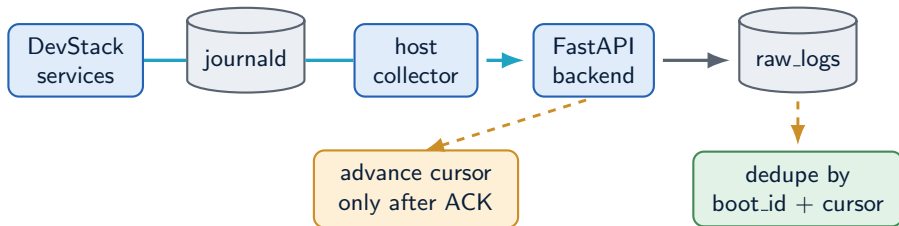
Central message: RCA Copilot converts noisy OpenStack logs into structured, bounded, explainable investigation evidence.

5. Current Implementation Status



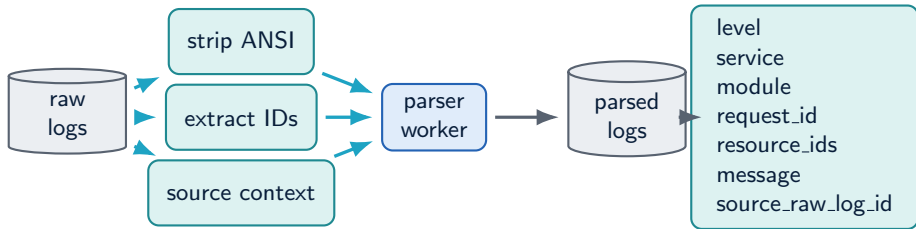
Solid boxes are implemented evidence construction. Dashed boxes are future explanation and operator-experience layers.

6. Ingestion: journald to MongoDB



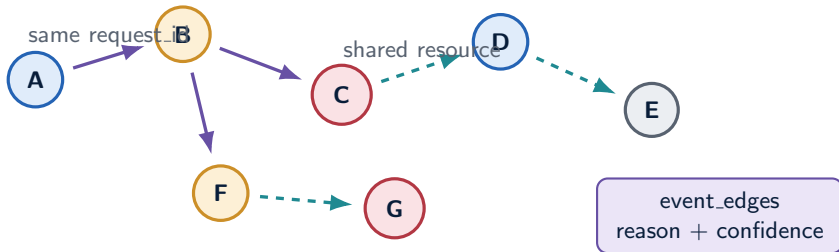
No guessing happens at ingestion. The raw evidence envelope is preserved before downstream interpretation.

7. Parsing: Raw Records to Structured Events



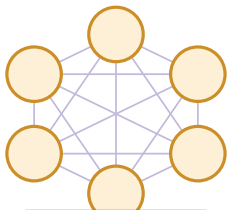
Parse failures are stored too. Missing structure becomes visible instead of silently disappearing.

8. Correlation Graph: Request and Resource Links



Edges are chronological correlation edges. They connect evidence; they do not prove causality.

9. Graph Safety: Why the Graph Does Not Explode



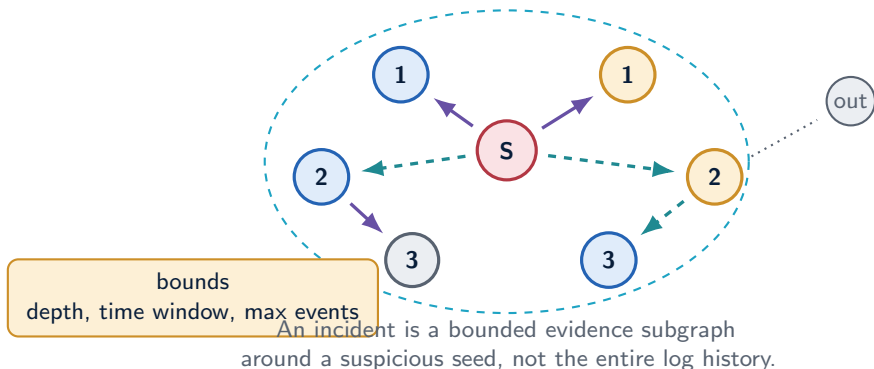
naive complete
 $O(n^2)$ graph



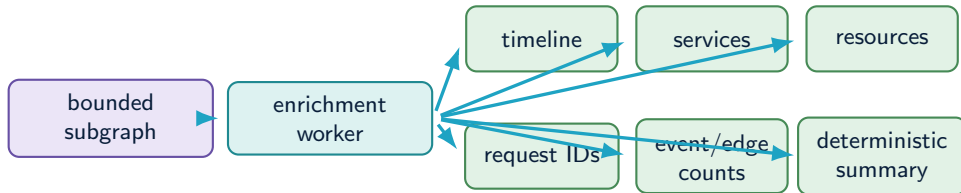
implemented: chronological consecutive edges
bounded groups, periodic groups skipped

This is both a performance control and a trust control: less false structure means clearer evidence.

10. Incident Construction: Seed Event to Bounded Subgraph

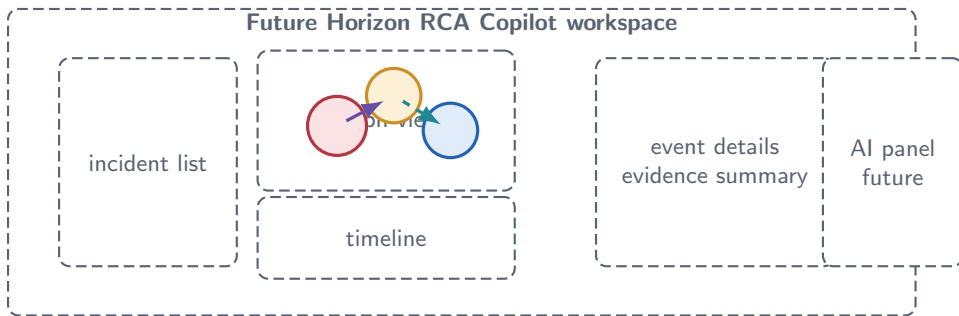


11. Enrichment: Deterministic Evidence Package



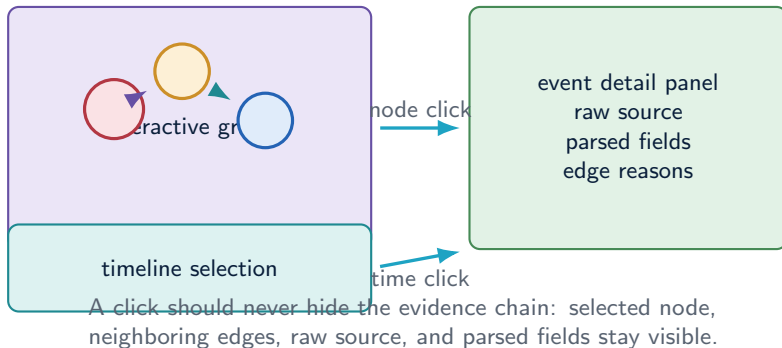
The evidence package is generated deterministically from stored events and edges before any AI layer exists.

12. Horizon Investigation Workspace

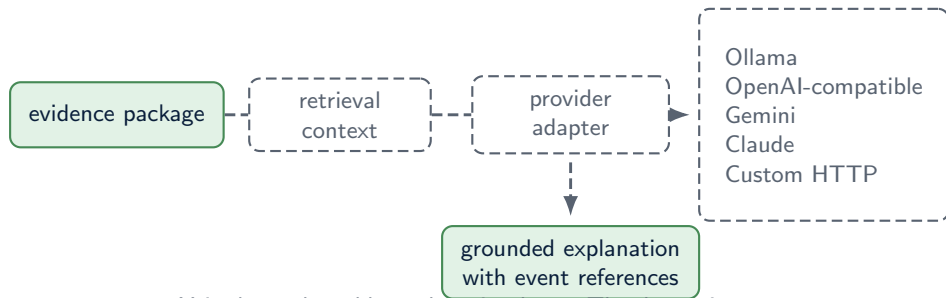


The UI moves between incident list, graph, timeline, raw event details, and visible evidence.

13. Dynamic Graph Interaction Concept

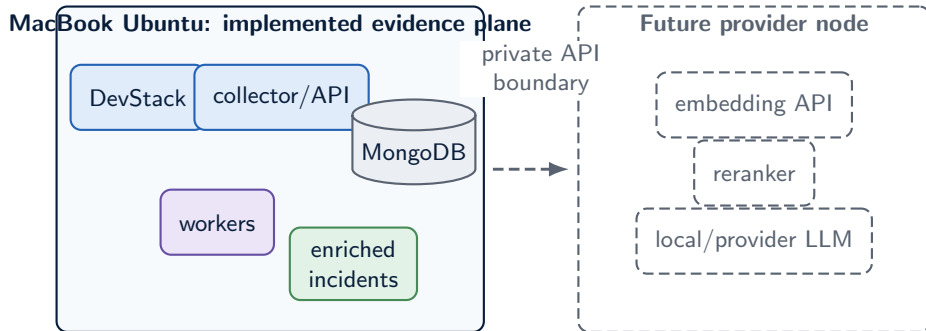


14. Provider-Agnostic AI/RAG Future

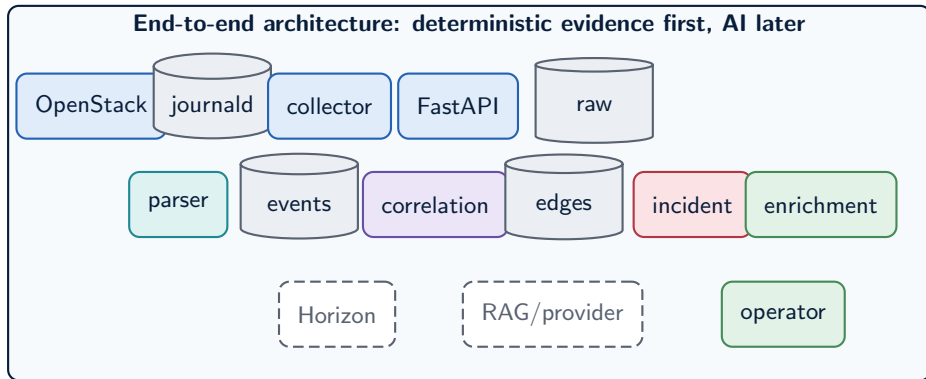


AI is the replaceable explanation layer. The deterministic evidence pipeline remains the source of truth.

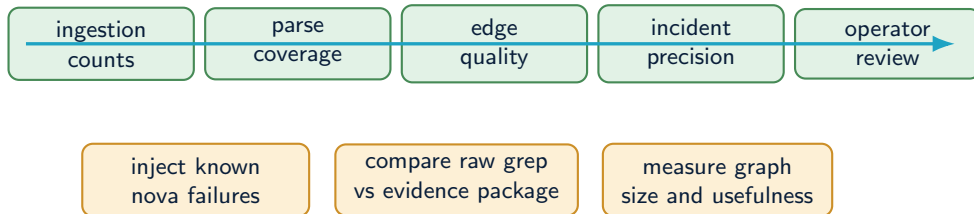
15. Two-Machine Architecture



16. Final Full Architecture



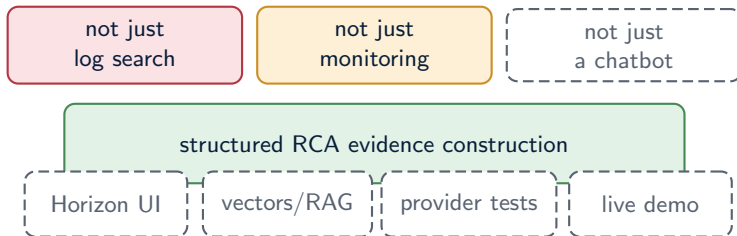
17. Validation Plan



Validation should prove that the system reduces investigation work without inventing causality or hiding raw evidence.

18. Roadmap and Final Takeaway

What is unique?



Final takeaway: RCA Copilot converts noisy OpenStack logs into structured, bounded, explainable investigation evidence.